GreenLight: Synthetic Patient Data for the Early Weeks of an Outbreak, Built from Past Diseases and Released Only When Ready

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Abstract

In the first weeks of a new outbreak, too few patients have known outcomes to train a reliable model of who dies, yet decisions and data sharing cannot wait. A natural idea is to borrow from earlier outbreaks, to say when the borrowed model is ready, and only then to release synthetic patient records. We build such a system, GreenLight, and evaluate it the way it would be used, on OutbreakReplay: a benchmark that replays 32 real outbreaks of hospitalised severe acute respiratory illness from Brazil's public surveillance (2.5 million patients, 2013–2025) day by day, never using information from the future. Four findings stand out. (1) Using each patient's final outcome from the day the record was entered, a common shortcut, inflates early accuracy enough to reverse a conclusion: on day 28 of COVID-19 in 2020 only 32% of the outcomes so used were known, and GreenLight's apparent advantage over real data alone disappeared once this was fixed. (2) In a pre-registered test on nine unseen seasons (2023–2025), simply pooling earlier seasons was *more* accurate than GreenLight and every adaptive borrowing method, and came within 0.001 AUC of a hindsight reference without any current data. (3) A stability-based readiness signal was honest (one borderline false signal in 31 outbreaks), whereas a standard minimum-sample-size rule declared readiness within 90 days in only 8 of 32 outbreaks, and never for other respiratory viruses. (4) Synthetic data released at that signal trained better models than the real records it was built from and than five standard generators trained on those records, because it carries borrowed knowledge. We report the negative result as found and release code to reproduce every number.

1 Introduction

When a respiratory pathogen causes a new wave of hospitalisations, the clinical question "who is likely to die?" arises long before the outbreak's own data can answer it. In the first weeks, few patients have been admitted and fewer have a known outcome. Models trained on these records alone are unstable, and the many COVID-19 prognostic models built from small early samples were mostly at high risk of bias (Wynants et al., 2020). Sharing patient records to pool evidence is slow, so synthetic patient data are often proposed as a stop-gap.

Three questions follow, and they are rarely studied together. *Can earlier outbreaks help?* Transfer from earlier diseases has worked for COVID-19 outcome prediction (Lichtner et al., 2021; Agarwal et al., 2022), and clinical-trial statistics offers a family of ways to borrow from historical data (Ibrahim & Chen, 2000; Hobbs et al., 2011; Schmidli et al., 2014; Kaizer et al., 2018). *When is a model ready?* Minimum sample sizes exist for models developed on one's own data (Riley et al., 2020), but not for a model that borrows. *Is synthetic data released early useful?* Generators trained on a few hundred records can only reproduce what those records contain.

We study all three on real data, under the conditions a health agency would face. Our contributions:

1. **OutbreakReplay, a time-respecting replay benchmark** over 32 outbreaks (influenza, other respiratory viruses and COVID-19) from public Brazilian surveillance, in which every method sees, each day, only the records entered so far and only the outcomes known so far (Section 3).

2. **GreenLight**, an interpretable system that votes on which earlier outbreaks the new one resembles, borrows from them with a strength that fades as its own data grow, falls back to a safe age rule when the pathogen is new, signals when its model is ready (it gives the *green light*), and only then releases labelled synthetic data (Section 4).

3. **A careful evaluation**: 9 borrowing baselines, 5 synthetic-data generators, a sample-size readiness rule, intervals at the patient and outbreak level, and a final test on nine seasons that were downloaded only after the plan, settings and analysis code were committed (Section 5).

Our main empirical findings are summarised in the abstract. Two are negative or cautionary: ignoring when outcomes become known can reverse conclusions, and on familiar seasons with a rich history, plain pooling beats adaptive borrowing. We think both are useful to anyone evaluating early-outbreak models.

2 Related work

Borrowing from historical data. Power priors (Ibrahim & Chen, 2000), commensurate priors (Hobbs et al., 2011), robust meta-analytic-predictive priors (Schmidli et al., 2014) and multi-source exchangeability models (Kaizer et al., 2018) decide how much to trust historical or supplementary data. GreenLight's borrowing rule is a simple member of this family; we include representatives of each as baselines.

Transfer and meta-learning for clinical prediction. Transfer learning (Pan & Yang, 2010) from non-COVID viral pneumonia (Lichtner et al., 2021) and from severe respiratory admissions (Agarwal et al., 2022) improved COVID-19 outcome prediction; MetaPred meta-learns across related clinical tasks (Zhang et al., 2019). Borrowing from the wrong source can hurt (negative transfer, Wang et al., 2019). These studies evaluate one fixed transfer at one time; we evaluate day by day, with a choice among many sources.

Readiness. Riley et al. (2020) give the minimum sample size for developing a prediction model with acceptable overfitting and precision. We compare it with our stability-based green light.

Real-time severity and reporting delay. Early case-fatality estimates are biased because recent cases have not yet resolved (Nishiura et al., 2009), and nowcasting corrects counts for reporting delay (Höhle & an der Heiden, 2014). We show the analogous problem for patient-level machine-learning replays.

Synthetic tabular data. CTGAN and TVAE (Xu et al., 2019), TabDDPM (Kotelnikov et al., 2023), adversarial random forests (Watson et al., 2023) and Bayesian networks, available in synthcity (Qian et al., 2023), are standard generators. Synthetic data are not automatically private (Stadler et al., 2022); our data are public, and we report a simple copy check.

3 Data and the OutbreakReplay protocol

Data. SIVEP-Gripe, Brazil's national surveillance of hospitalised severe acute respiratory illness, is published by the Ministry of Health under CC-BY (Ministério da Saúde do Brasil, 2026). We use hospitalised patients with a recorded outcome (cure or death), the H1N1 pandemic of 2009 and the years 2013–2025. An *outbreak* is one file year and laboratory class: influenza, other respiratory virus, or COVID-19 (from 2020). Features are those the admission form collects in every year: age, sex, self-reported race, state, six symptoms and ten conditions (yes / no / missing). The development set has 24 outbreaks (2009–2022; 2.14 million patients), 23 of which have an earlier outbreak to borrow from; the frozen test set has 9 (2023–2025; 0.37 million). Table 1 describes each outbreak.

Table 1: Outbreaks. Patients: hospitalised with a recorded outcome. Missing: share of the 16 yes/no fields not collected or not filled in. Day 0: start of the replay. Test: patients admitted on days 91–182 (July 2020 for COVID-19 2020). Green: GreenLight's green-light day (−: none by day 90). H1N1 2009 is used only as an earlier outbreak to borrow from.

Set	Outbreak	Patients	Died (%)	Median age	Missing (%)	Day 0	Test patients	Test deaths	Green
Development	Influenza 2013	6,146	16.2	35	7	2013-04-02	2,483	371	70
	Other virus 2013	3,120	7.1	1	6	2013-04-10	1,046	85	–
	Influenza 2014	1,762	18.6	36	8	2014-04-01	751	135	46
	Other virus 2014	2,249	5.7	1	5	2014-03-25	792	33	34
	Influenza 2015	1,257	16.2	39	6	2015-03-09	579	96	36
	Other virus 2015	2,418	5.0	1	6	2015-02-23	1,129	60	64
	Influenza 2016	11,824	18.8	39	6	2016-02-12	3,896	849	32
	Other virus 2016	4,607	6.6	1	4	2016-03-07	2,412	154	70
	Influenza 2017	3,255	15.2	44	4	2017-02-01	1,577	225	24
	Other virus 2017	4,689	6.4	1	4	2017-02-09	2,382	139	46
	Influenza 2018	7,856	17.3	36	4	2018-03-01	3,973	746	44
	Other virus 2018	8,071	5.8	1	3	2018-01-19	4,165	239	80
	Influenza 2019	6,460	17.7	36	36	2019-03-11	2,736	470	12
	Other virus 2019	8,806	5.7	1	46	2019-03-08	2,264	132	22
	COVID-19 2020	645,001	33.2	61	42	2020-02-26	93,626	31,426	54
	Influenza 2020	2,336	13.4	34	43	2020-01-09	397	75	34
	Other virus 2020	4,620	7.5	3	46	2020-01-02	1,119	112	38
	COVID-19 2021	1,110,621	32.7	57	46	2021-01-01	507,241	165,882	28
	Influenza 2021	10,012	15.5	59	50	2021-04-07	202	26	38
	Other virus 2021	19,629	3.0	1	57	2021-01-08	4,331	147	50
	COVID-19 2022	214,982	27.7	70	43	2022-01-01	36,955	8,363	30
	Influenza 2022	10,826	12.6	49	45	2022-01-01	1,648	176	30
	Other virus 2022	31,050	2.6	1	59	2022-01-02	12,564	353	42
Frozen test	COVID-19 2023	44,317	19.7	68	42	2023-01-01	10,114	2,165	28
	Influenza 2023	11,196	9.6	11	50	2023-01-08	7,410	747	40
	Other virus 2023	46,180	2.1	1	59	2023-01-01	18,583	366	52
	COVID-19 2024	27,720	18.6	68	40	2024-01-01	3,592	717	34
	Influenza 2024	22,836	10.4	34	46	2024-01-03	10,798	1,179	24
	Other virus 2024	61,184	2.4	1	57	2024-01-02	28,431	671	38
	COVID-19 2025	15,796	16.3	61	38	2025-01-01	1,648	318	32
	Influenza 2025	41,518	11.5	47	41	2025-01-01	24,959	3,359	18
	Other virus 2025	97,269	2.6	1	54	2025-01-01	43,737	1,007	58
Source only	H1N1 2009	21,123	9.5	24	47	–	–	–	–

Replay. Each outbreak is replayed from day 0 to day 90 in steps of 2 days. Day 0 is the first record (COVID-19) or the first week with at least 20 records (seasonal outbreaks). On day d a method may use (i) records of the new outbreak (we call it *fob*) entered by day d, (ii) of those, only outcomes *known* by day d, and (iii) earlier outbreaks' records and outcomes known before fob's day 0. An outcome counts as known from the later of the entry date and the outcome date (falling back to the case-closure date). Each day's model is scored by AUC on the patients admitted on days 91–182 (July 2020 for COVID-19 in 2020), which no method sees. Two references: *gold*, the same model class fitted in hindsight on all of fob's other records, and *rank by age*. The *gap to gold* is gold's AUC minus the method's.

4 GreenLight

Kinship Vote. Each week, every earlier outbreak predicts fob's newly admitted patients: who is hospitalised (age band, sex and the yes/no fields fob's form collects) and who dies (its severity model with fob's own death rate). Predictions are scored by log-likelihood, prequentially (Dawid, 1984; Gneiting & Raftery, 2007). A *Stranger* candidate predicts from fob's own earlier weeks. Vote shares are a softmax of accumulated scores with sharpness $\tau = 0.01$, starting from equal shares per pathogen group.

Handover Rule. The severity model is a MAP logistic regression whose prior mean is the vote-weighted average of the earlier outbreaks' coefficients and whose strength is $\lambda = 10\,(1 - s)$, where s is the Stranger's share; the intercept (fob's own death rate) and the state effects are never borrowed. As fob's records accumulate, the fixed prior is outweighed and fob's own data take over.

Lab-Label Fallback. If the laboratory class of fob has no earlier outbreak, GreenLight uses a one-parameter age trend until 20 deaths have been scored on fresh data, then borrows.

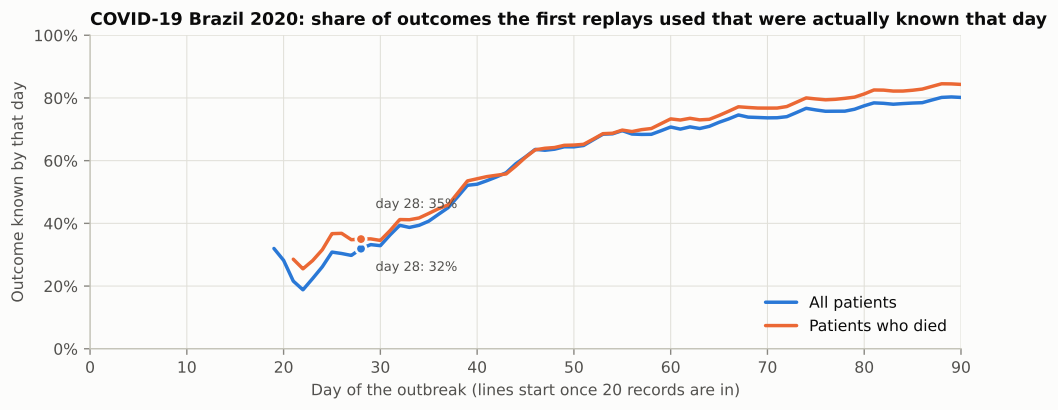


Figure 1: COVID-19, Brazil 2020: share of the outcomes used by an "outcome on entry" replay that were actually known on that day.

The green light. GreenLight declares its model ready on day d if at least 10 deaths have a known outcome, the rank correlation of its risk scores today with those of 8 days earlier is at least 0.99 on two consecutive checks, and GreenLight is not on the fallback. These two values were re-chosen on the development set after the outcome-timing fix (Section 5.1).

Synth Release. After the green light, GreenLight generates synthetic patients: profiles from a dependency chain of logistic models (each yes/no field given age band, sex and earlier fields, with priors borrowed as above) and death drawn from GreenLight's risk model. Every row is labelled synthetic with its provenance.

Glass Box Report. An HTML page per outbreak and day shows the vote, the risk table with borrowed shares, the green light conditions and synthetic-versus-real comparisons; examples are in the code repository.

5 Experiments and results

All settings were chosen on the development set. Intervals are 95% percentile bootstrap intervals: over outbreaks (2,000 resamples, paired where methods are compared) or over test patients (1,000 stratified resamples).

5.1 Outcome timing reverses a conclusion

Our first replays, like many retrospective evaluations, used each patient's final outcome from the day the record was entered. For COVID-19 in 2020, on day 28 only 32% of the outcomes so used, and 35% of the deaths, had actually occurred and been recorded (Figure 1). After the fix, the day GreenLight came within 0.02 AUC of gold moved from 28 to 36, the same as real data alone; GreenLight's apparent COVID-19 advantage was an artefact. On familiar seasons GreenLight still matched gold from day 0. The fix also produced two false green lights under the original readiness settings, which we therefore re-chose (Section 5.3).

5.2 Who dies: borrowing versus baselines

Figure 2 shows typical replays. Table 2 compares GreenLight with nine baselines, all using the same design and data: a pooled model of earlier outbreaks (no fob data); stacking the pooled earlier outbreaks with fob's records; fine-tuning towards the pooled model; GreenLight without the vote (equal weights); a meta-analytic prior and its robust version (Schmidli et al., 2014); a borrowing strength chosen by marginal likelihood (commensurate-style, Hobbs et al., 2011); source averaging by marginal likelihood (a simplification of multi-source exchangeability, Kaizer et al., 2018); and a Reptile meta-learned starting point (Nichol et al., 2018) in the spirit of MetaPred (Zhang et al., 2019).

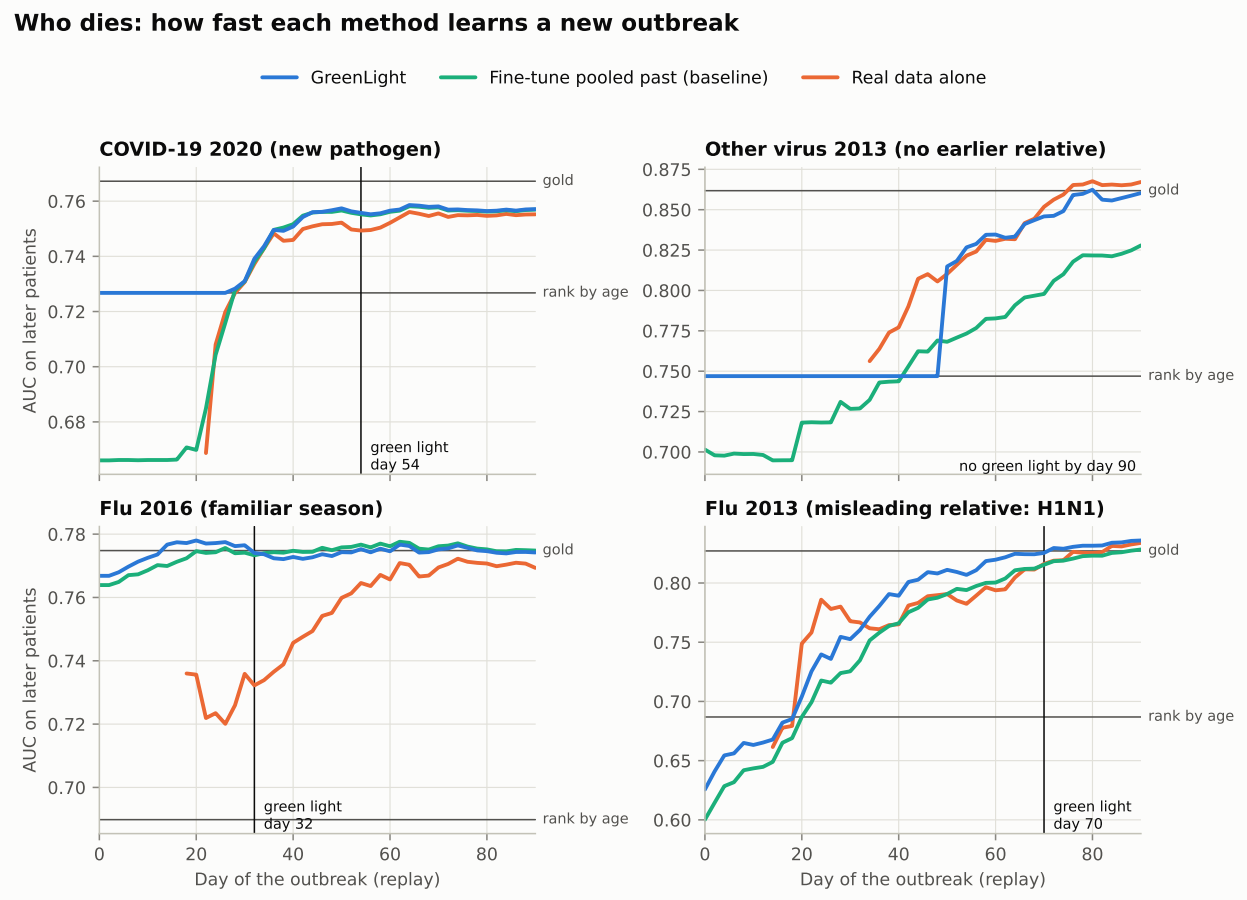


Figure 2: AUC on later patients by replay day for four development outbreaks.

Table 2: Mean gap to gold over days 0–90 and its difference from GreenLight (positive: GreenLight better), with 95% paired bootstrap intervals over outbreaks. Development: 23 outbreaks (2013–2022). Frozen test: 9 outbreaks (2023–2025).

	Development (23)		Frozen test (9)	
Method	gap	minus GreenLight	gap	minus GreenLight
GreenLight	0.0052	–	0.0067	–
Real data alone	0.0520	+0.047 [0.028, 0.065]	0.0393	+0.033 [0.020, 0.046]
Pooled earlier outbreaks (no fob data)	0.0210	+0.016 [0.003, 0.032]	0.0010	−0.006 [−0.010, −0.002]
Stacking	0.0101	+0.005 [−0.002, 0.013]	0.0006	−0.006 [−0.010, −0.002]
Fine-tuning	0.0085	+0.003 [−0.001, 0.008]	0.0070	+0.000 [−0.001, 0.001]
GreenLight without the vote	0.0087	+0.004 [0.001, 0.006]	0.0074	+0.001 [−0.000, 0.002]
Source averaging	0.0081	+0.003 [0.001, 0.005]	0.0082	+0.002 [0.001, 0.002]
Meta-analytic prior	0.0140	+0.009 [0.004, 0.016]	0.0097	+0.003 [0.002, 0.004]
Robust meta-analytic prior	0.0166	+0.011 [0.006, 0.019]	0.0099	+0.003 [0.002, 0.004]
Empirical-Bayes strength	0.0146	+0.009 [0.005, 0.014]	0.0086	+0.002 [0.000, 0.004]
Meta-learned start	0.0124	+0.007 [0.005, 0.010]	0.0130	+0.006 [0.003, 0.009]

On the development set, GreenLight had the lowest gap, but its margin over stacking and fine-tuning was not distinguishable from zero; the vote's own contribution (GreenLight versus equal weights) was small but consistent. Every method that borrows was far better than real data alone. The Lab-Label Fallback

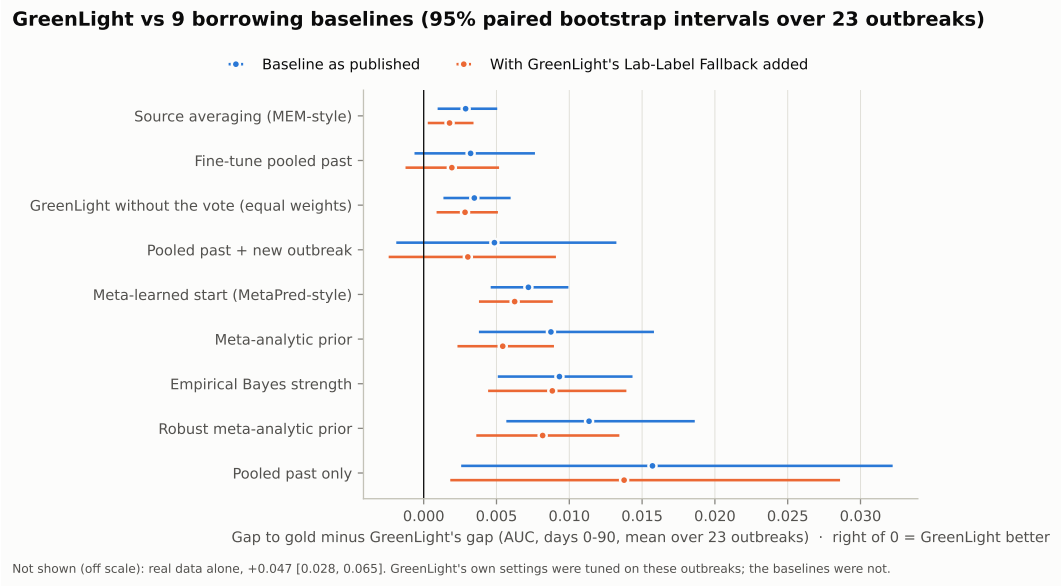


Figure 3: Development set: gap to gold of each baseline minus GreenLight's, with and without GreenLight's Lab-Label Fallback.

Table 3: Calibration on the green light day, scored on the later test patients: mean over outbreaks [95% bootstrap interval]. O/E: observed / expected deaths (1 = right level). Slope: logistic recalibration slope (1 = right spread).

	Development (22)			Frozen test (9)		
Model	O/E	Slope	Brier	O/E	Slope	Brier
GreenLight	0.94 [0.84, 1.03]	0.96 [0.90, 1.01]	0.097	1.15 [1.05, 1.29]	0.97 [0.92, 1.02]	0.081
Real data alone	1.08 [0.93, 1.26]	0.75 [0.67, 0.84]	0.105	1.08 [0.97, 1.20]	0.84 [0.75, 0.92]	0.084
Pooled earlier outbreaks	0.80 [0.73, 0.87]	0.89 [0.81, 0.96]	0.102	0.70 [0.58, 0.83]	1.04 [0.99, 1.09]	0.082
Stacking	1.00 [0.90, 1.08]	0.92 [0.87, 0.97]	0.098	1.15 [1.03, 1.29]	1.04 [0.99, 1.09]	0.080
Gold (hindsight)	1.02 [0.97, 1.08]	1.29 [1.16, 1.42]	0.096	1.09 [1.03, 1.14]	1.10 [1.06, 1.15]	0.080

mattered most for safety: without it, every borrowing baseline ranked COVID-19 patients worse than age alone on 24–92 of the first 90 days of 2020, against none for GreenLight (Figure 3).

Frozen test. The plan (seasons, settings, endpoints and success criteria) was committed before the 2023–2025 files were downloaded. Five of six criteria held: borrowing beat real data alone; GreenLight never ranked worse than age; at most one false green light; GreenLight synthetic data beat the real records; no clear gain over fine-tuning. The sixth failed in the negative direction: **stacking, and even the pooled earlier seasons with no current data, were more accurate than GreenLight** (gaps 0.0006 and 0.0010 against 0.0067; GreenLight better in 1 of 9). By 2023 the history holds many seasons of each pathogen, and season-to-season severity patterns are stable; GreenLight's fixed prior strength lets a few dozen early records outweigh this history too soon. We did not re-tune GreenLight.

Calibration. AUC measures ranking only. Table 3 reports, on the green light day, whether predicted risks match what happened to the later patients (Van Calster et al., 2019). GreenLight's ranking was well calibrated (slope near 1, unlike real data alone, whose risks were too extreme). Its overall level was right on average in development but under-predicted deaths by about 15% on the frozen test; stacking did the same and even the gold model under-predicted by 9%, so later patients in those seasons died somewhat more often than earlier ones. The pooled model, which ignores fob's own death rate, over-predicted deaths by 20–30%. Brier scores of GreenLight, stacking and gold were within 0.002 of each other.

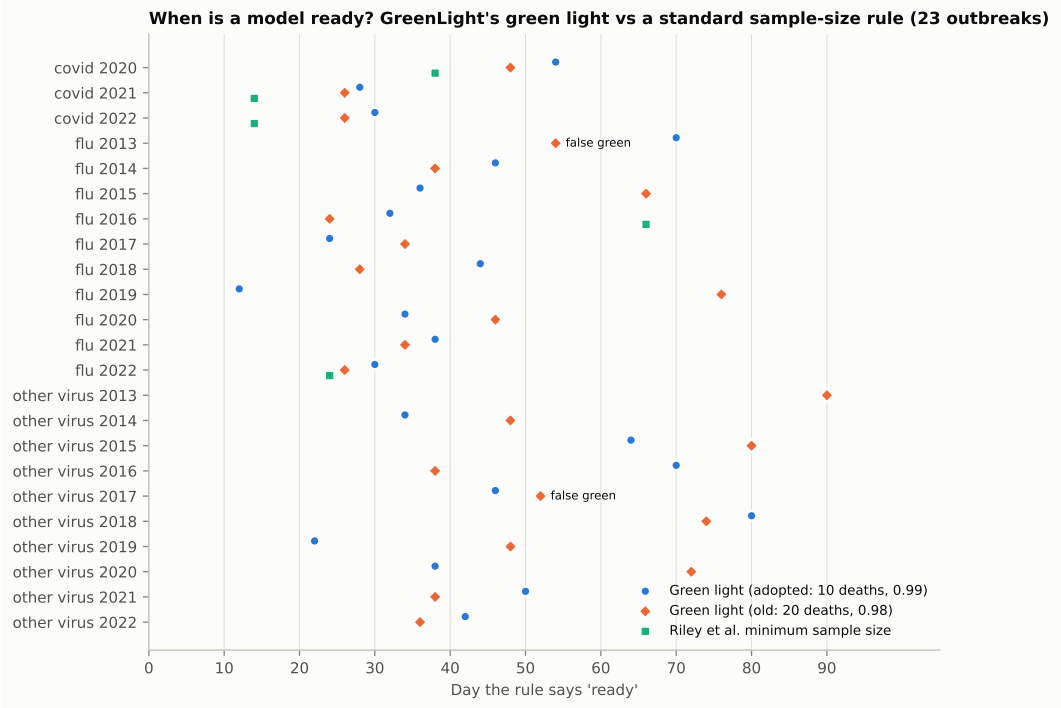


Figure 4: Day each readiness rule declares a model ready, development outbreaks.

5.3 Is the model ready?

With the adopted settings (10 deaths, stability 0.99), 22 of 23 development outbreaks turned green by day 90 (median day 38, interval [32, 46]) with no false signal (a model more than 0.02 AUC below gold), and 9 of 9 frozen test outbreaks (median day 34) with one borderline false signal (other virus 2023, gap 0.021). Choosing the settings with each development outbreak held out gave one false signal in 22. At the patient level, 22 of 31 green days were firmly correct, 9 uncertain because small test windows give wide intervals, and none firmly false (Figure 4). Riley's minimum sample size (Riley et al., 2020), with the anticipated R^2 taken from the pooled earlier outbreaks, required 2,100–7,100 patients with a known outcome: it declared readiness in 8 of 32 outbreaks (the six COVID-19 seasons and influenza 2016 and 2022) and never for other respiratory viruses, although for most seasons the borrowed model is near gold from day 0.

5.4 Which outbreaks does the new one resemble?

The vote put most weight on the right pathogen group for 19 of 21 development outbreaks with an earlier group member in weeks 1–4 and 20 of 21 from week 8; on the frozen test, 9 of 9 in every week checked. When the group share was at least 95%, it was right 97.6% of the time. The Stranger took over for COVID-19 in 2020 in week 4, after 3,538 patients (Figure 5); it needs hundreds of patients, so for small outbreaks the laboratory label is the practical safeguard.

5.5 Synthetic data

On the green light day we trained a ridge logistic model on 50,000 synthetic rows and scored it on the later real patients (Figure 6). Development: GreenLight synthetic data 0.796 [0.781, 0.811]; the real records it was built from 0.730; ARF 0.770, TVAE 0.704, bootstrap resampling 0.679, Bayesian network 0.676, CTGAN 0.613, each trained on those same records. For outbreaks with fewer than 1,000 records, adding GreenLight synthetic rows to the real ones raised AUC by 0.086 [0.064, 0.109]. Frozen test: GreenLight minus real records +0.035 [0.013, 0.058], minus ARF +0.016 [0.008, 0.024]. TabDDPM (Kotelnikov et al., 2023) could

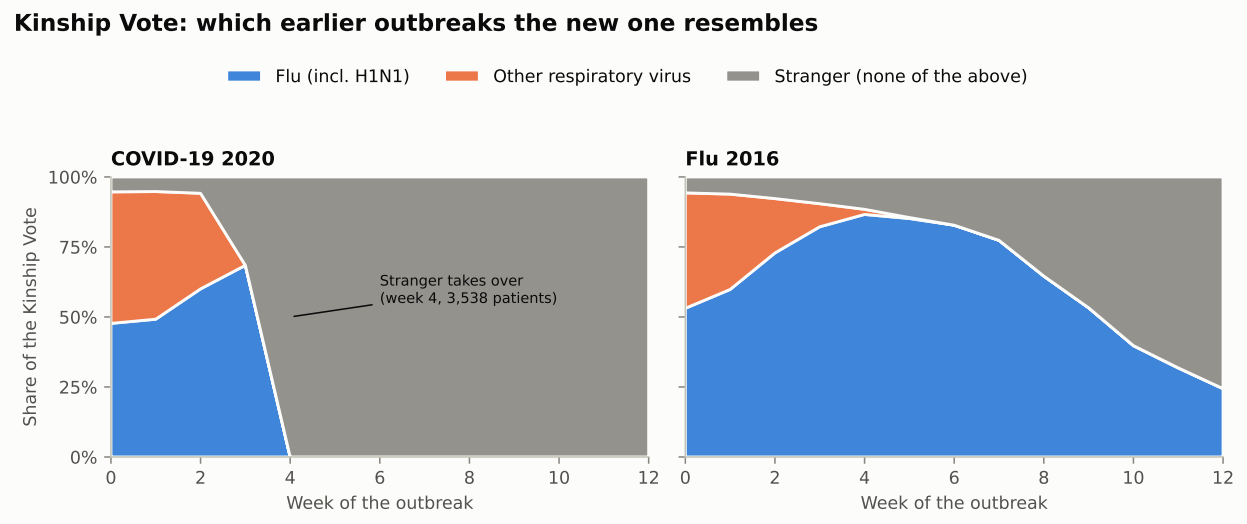


Figure 5: Kinship Vote shares by pathogen group and the Stranger.

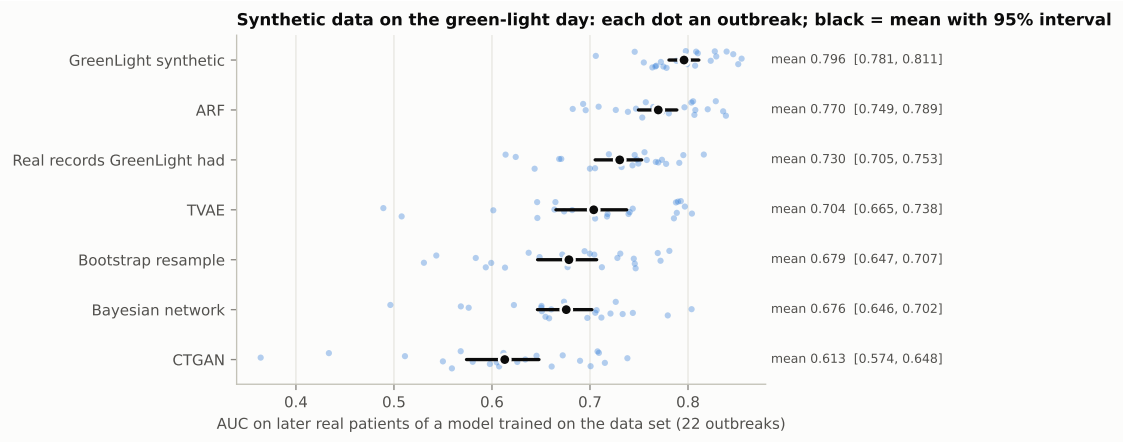


Figure 6: AUC on later real patients of a model trained only on each data set, development outbreaks.

not be run on our CPU budget. The comparison is not like for like by design: the generators see only fob's records, while GreenLight's data carry borrowed knowledge. The gap shrinks where fob's own records are plentiful, and in one season (influenza 2022) bootstrap and ARF beat GreenLight.

Copy check. Records are coarse (age in years, sex, race, state, 16 yes/no fields, outcome), so a genuinely new patient often matches an earlier one exactly. We therefore compare the share of synthetic rows that exactly match a record the generator saw with the share of later, unseen real patients that do. GreenLight's synthetic data matched at the chance level: 12.7% against 11.4% (difference +0.013 [−0.009, 0.035]) on the development set and 11.3% against 13.5% (−0.022 [−0.052, 0.000]) on the frozen test. Bootstrap resampling, which copies by design, matched 100%. This is a check for copying, not a privacy guarantee (Stadler et al., 2022); the source data are public.

Sensitivity to outcome timing. Counting an outcome only from the case-closure date, a more cautious choice, weakened GreenLight slightly on the development set (mean gap 0.0068 against 0.0053; 94 days below age against 66) and gave one false green light in 22 (other virus 2017; median green day 41 against 38). For COVID-19 in 2020, real data alone came within 0.02 of gold two days before GreenLight (day 34 against 36). The conclusions do not change.

6 Discussion

What GreenLight contributes. Not a better borrowing formula: plain pooling is a strong, simple competitor and was better on the frozen test. What held up on unseen seasons is what the baselines lack: a safe fallback for a new pathogen, a readiness signal that was honest, a vote that found the right group, and synthetic data whose usefulness comes from borrowed knowledge rather than from the generator.

Why pooling won. GreenLight lets fob's data take over at a fixed rate. When the history is rich and the new season familiar, that hands over too soon. A natural fix, letting the borrowing strength grow with the vote's confidence that fob is familiar, is new development work and needs a new held-out test.

Limitations. One country and one surveillance system for the main analyses; one truly new pathogen (COVID-19); admission-form features only; AUC on a later window as the main metric; a fixed ridge model to score synthetic data. An earlier out-of-country check on Mexico's COVID-19 data used outcome-on-entry timing because survivors have no outcome date there, and is not reported as evidence. The test outbreaks contain no new pathogen.

Broader impact. Synthetic records here hold no more information than the real records and the earlier outbreaks they come from. They are labelled synthetic, must not be presented as real patients, and cannot reveal a risk factor nobody has recorded.

7 Reproducibility

All data are public (CC-BY). Code, pinned environments, configuration files, seeds, the frozen-test plan and a running results log are available at [PENDING: anonymous code link]. The frozen-test plan was committed before the test data were downloaded; the commit hash is given in the repository.

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